

TAJAMUL ASHRAF

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RESEARCH INTERESTS

Multimodal Agents, Vision-Language Reasoning, Medical AI, Video Understanding, Embodied AI, Visual Grounding, Domain Generalization, and AI for Science

EDUCATION

Indian Institute of Technology Delhi

Aug 2022 – June 2024

Master's (Research) in Computer Science, GPA: 9.28/10.0

New Delhi, India

Advisor: **Prof. Chetan Arora**

Thesis: *Domain Adaptation in Breast Cancer Detection from Mammograms.*

National Institute of Technology Srinagar

Aug 2018 – July 2022

B.Tech. in Information Technology, GPA: 8.34/10.0

Srinagar, India

Advisor: **Prof. Janibul Bashir**

Thesis: *Robust DL Models for Analyzing the Impact of Coherent Climatic Factors.*

SELECTED PUBLICATIONS

 Best Paper Award, * equal contribution, † my role as mentor

- J2** Komal Kumar*, **Tajamul Ashraf***, Omkar Thawakar, Rao Muhammad Anwer, Hisham Cholakkal, Mubarak Shah, Ming-Hsuan Yang, Phillip Torr, Salman Khan, Fahad Khan, “*LLM Post-Training: A Deep Dive into Reasoning Large Language Models.*” In **IEEE TPAMI**. [📄 Paper](#), [🔗 Code](#)
- C9** Rozain Shakeel, Abdul Rahman, Muneeb Mushtaq, Tausifa Jan Saleem, **Tajamul Ashraf†**, “*MedSPOT: A Workflow-Aware Sequential Grounding Benchmark for Clinical GUI.*” In **ECCV’26**. [🔗 Code](#) [📄 Paper](#)
- J1** **Tajamul Ashraf**, Rajes Manna, Partha Purkayastha, Tawaheed Tariq, Janibul Bashir, “*Context Aware Grounded Teacher for Source Free Object Detection.*” In **IJCV**. [🔗 Code](#) [📄 Paper](#)
- C8** Akash Ghosh, **Tajamul Ashraf†**, Rishu Singh, Numaan Saeed, Sriparna Saha, Xiuying Chen, Salman Khan, “*CarePilot: A Multi-Agent Framework for Long-Horizon Computer Task Automation in Health-care.*” In **CVPR’26 Findings**. [🔗 Code](#) [📄 Paper](#)
- C7** **Tajamul Ashraf***, Amal Saqib*, Jean Lahoud, Hisham Cholakkal, Mubarak Shah, Philip Torr, Fahad Khan, Rao Anwer, Salman Khan, “*Agent-X: Evaluating Deep Multimodal Reasoning in Vision-Centric Agentic Tasks.*” In **ICLR’26**. [🔗 Code](#) [📄 Paper](#) [📊 Data](#)
- C6** **Tajamul Ashraf**, Janibul Bashir, “*TITAN: Query-Token based Domain Adaptive Adversarial Learning.*” In **ICCV 2025**. [🔗 Code](#) [📄 Paper](#)
- C5** Jinhong Wang*, **Tajamul Ashraf*†**, Zongyan Han, Jorma Laaksonen, Rao M. Anwer, “*MIRA: A Novel Framework for Fusing Modalities in Medical RAG.*” In **ACM Multimedia 2025**. [🔗 Code](#) [📄 Paper](#)
- C4** Bhuvan Sachdeva, Naren Akash, **Tajamul Ashraf**, Simon Mueller, Thomas Schultz, Maximilian W. M. Wintergerst, Niharika S. Prasad, Kaushik Murali, Mohit Jain, “*Phase-Informed Tool Segmentation for Manual Small-Incision Cataract Surgery.*” In **MICCAI 2025**. [📄 Paper](#)
- C3** **Tajamul Ashraf**, Krithika Rangarajan, Mohit Gambhir, Richa Gauba, Chetan Arora, “*D-MASTER: Mask Annealed Transformer for Unsupervised Domain Adaptation in Breast Cancer Detection from Mammograms.*” In **MICCAI 2024**. [🔗 Code](#) [📄 Paper](#)
- C2**  **Tajamul Ashraf**, Tisha Madame, “*HF-Fed: Hierarchical Based Customized Federated Learning Framework for X-Ray Imaging.*” In **MICCAI 2024 Workshop (Deep-Brea³th)**. [🔗 Code](#) [📄 Paper](#)
- C1** **Tajamul Ashraf**, Fuzayil Mir, Iqra Altaf Gillani, “*TransFed: A Way to Epitomize Focal Modulation Using Transformer-Based Federated Learning.*” In **WACV 2024**. [🔗 Code](#) [📄 Paper](#)

RESEARCH EXPERIENCE

Nanyang Technological University (NTU)

Mar 2026 – Present

Research Manager Host: Prof. Adams Wai Kin Kong

Nanyang, Singapore

- Collaborating with the **Interpol** team on video understanding and spatiotemporal reasoning for surveillance applications.
- Developing multimodal agentic systems for long-horizon temporal reasoning and decision making.

King Abdullah University of Science and Technology (KAUST)

Dec 2025 – Mar 2026

Research Consultant Host: Prof. Bernard Ghanem

Remote

- Introduced **MedCTA**, a benchmark for evaluating multimodal clinical tool agents.
- Developed evaluation protocols for grounded and reliable tool-augmented clinical reasoning.

Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI)

Jan 2025 – Nov 2025

Research Engineer Hosts: Prof. Salman Khan, Prof. Rao Anwer

Abu Dhabi, UAE

- Curated **Agent-X**, a large-scale benchmark for perception, grounding, and decision-making in real-world agentic tasks; Awarded **2nd Place** (Research Track) at the Agentic AI Summit, UC Berkeley (2025).
- Built a multimodal RAG framework (**MIRA**) for factual and adaptive medical agents.
- Authored a survey on post-training techniques for reasoning LLMs.

Microsoft Research (MSR)

July 2024 – Dec 2024

Research Intern Hosts: Dr. Mohit Jain, Dr. Prachi Jain

Bangalore, India

- Built LLM-based Knowledge-Graph with RAG framework on a semi-structured KB for medical chatbots.
- Jointly introduced *Cataract-MSICS* dataset and proposed a phase-conditioned method with pseudo-labeling for surgical tool segmentation.

AI & Robotics Technology Park (ARTPARK), I-Hub

Dec 2021 – May 2022

Visiting Student Host: Prof. Raghu Krishnapuram

Bangalore, India

- Developed an automated washroom-cleaning robot *RoboJanitor*; implemented pose estimation, detection, classification, and SLAM.
- Built 3D pose models and REST APIs for production deployment.

Indian Institute of Science (IISc)

Mar 2021 – Sep 2021

Research Intern Host: Prof. S. N. Omkar

Bangalore, India

- Built a human-pose comparison model to assist yoga instruction in live online settings.
- Used BlazePose for keypoint extraction and delivered a feedback mechanism for real-time pose correction.

ACADEMIC SERVICES

Teaching Assistant

Probability and Statistics (SC2000), NTU Singapore (Spring 2026), instructed by **Prof. Adams Kong**.

Trustworthy Medical Vision (CV8502), MBZUAI (Fall 2025), instructed by **Prof. Yutong Xie**.

Computer Vision for Robotics (JRL780), IIT Delhi (Spring 2024), instructed by **Prof. Chetan Arora**.

Computer Science Fundamentals and Programming (COL100), IIT Delhi (Fall 2023), instructed by **Prof. Chetan Arora** and **Prof. Vireshwar Kumar**.

Data Structures and Algorithms (COL106), IIT Delhi (Fall 2022), instructed by **Prof. Naveen Garg** and **Prof. Ashish Chiplunkar**.

Reviewer

2026	NeurIPS, ACM Multimedia, MICCAI, IJCV
2025	ICCV, ACM Multimedia, MICCAI, WACV, Pattern Recognition, IJCV, TAI
2024	MICCAI, WACV, CVIP, TPDS
2023	CVIP, HPEC

Volunteering

Program Committee Member: ACM Multimedia 2026 (Rio de Janeiro, Brazil).

Volunteer: ICLR 2026 (Rio de Janeiro, Brazil), ACM COMPASS 2025 (Toronto, Canada), CVIP 2023 (IIT Jammu, India).

Invited Talks

Invited speaker at *Sangam 2.0* (NIT Srinagar Alumni Meet), delivering a talk on *Research Directions in current AI landscape*, covering research opportunities, higher studies, and emerging trends in AI (2025).

Delivered a keynote talk on “Multimodal Systems for Vision-Language Domains in Advancing Healthcare” at the BRAIN-AID Seminar, University of Lagos, Nigeria (April 2025)

Gave two talks on research directions, placements, and internships at NIT Srinagar (June 2024)

Mentorship

To give back, I co-founded Gaash Research Lab to foster research culture among students and actively mentor Bachelor’s and Master’s students on thesis projects and publication-oriented research. Non-exhaustive list:

2026

14. Ilakkiya Selvakumar (NTU Singapore) *Benchmarking Multimodal Agents for Surveillance Applications*
13. Uzair Nazir, Abdul Muizz (NIT Srinagar) *OCR and kashmir language translation*
12. Adab Ismail (NIT Srinagar) *Vision language modeling for object tracking*
11. Sheikh Samee ul Haq, Aabid Nabi (NIT Srinagar) *Robust Tool Segmentation in Surgical Scenarios*
10. Rozain Shakeel, Muneeb Ahmad Ganie (NIT Srinagar) *Grounding and Reasoning for Clinical GUI*
9. Tafheem Malik, Iram Naseer (NIT Srinagar) *Multimodal Image and Video Generation using Reasoning*
8. Ifrah Mushtaq (University of Kashmir) *ASR and Phonetic Analysis for Kashmiri Language*

2025

7. Burhaan Rasheed Zargar (NIT Srinagar) *Script-Aware Flow Matching for Kashmiri Text-to-Speech*
5. Abrar Riyaz (NIT Srinagar), Wasif Tak (Thapar University) *Language-Conditioned Surgical Tool Segmentation*
4. Tawaheed Tariq, Sonia Yadav (NIT Srinagar) *Query-Driven Reasoning for Multi-modal MOT*

2024

3. Suhaib Salmani (NIT Srinagar) *VLM-based Masked Learning for Efficient Medical Object Detection*
2. Rajes Manna (NIT Srinagar) *Context Aware Grounded Teacher for Source-Free Object Detection*

2023

1. Asrar ul Haq (NIT Srinagar) *Transformer-Based Unsupervised Domain Adaptation in a Federated Setup*

ACHIEVEMENTS AND HONORS

- **ICLR 2026 Registration and Travel Grant** (Rio de Janeiro, Brazil).
- **MICCAI 2025 Outstanding Reviewer Award.**

- Awarded travel and registration grants for **ICCV 2025** (Honolulu, Hawaii), **ACM Multimedia 2025** (Dublin, Ireland), and the **Agentic AI Summit 2025** (UC Berkeley, California).
- **Second Place** in the Agent-X Competition at the Agentic AI Summit 2025, UC Berkeley.
- Secured 2nd place in the MBZUAI Department Logo Design Competition.
- Awarded a \$500 compute grant from **alphaXiv** for contributions to LLM research.
- **Best Student Paper Award**, Deep-Brea³th Workshop, MICCAI 2024 (HF-Fed).
- Nominated as a **Diversity Intern** by Microsoft Canada.
- Awarded the **IIT Delhi Endowment Grant** and travel grant for **MICCAI 2024** (Marrakesh, Morocco).
- Gold **Microsoft Student Learn Ambassador** at NIT Srinagar.
- Won India's largest **National Entrepreneurship Challenge** at IIT Bombay (100,000 INR cash prize).

MEDIA APPEARANCES

- *A New Stress Test for AI Agents That Plan, Look, and Click.* (2025) [\[Link\]](#)
- *Meet a Microsoft Learn Student Ambassador Graduate: Tajamul Ashraf.* (2022) [\[Link\]](#)
- *NIT Srinagar Students Win India's Largest National Entrepreneurship Challenge.* (2022) [\[Link\]](#)
- *Hult Prize: A Futuristic Approach to Entrepreneurship.* (2022) [\[Link\]](#)
- *Workshop on Entrepreneurship, Research and Innovation at NIT Srinagar.* (2021) [\[Link\]](#)

SKILLS & INTERESTS

Technical Skills	Python, C/C++, PyTorch, DeepSpeed, FSDP, Model/Sequence Parallelism
Leadership / Outreach	Founder of Ralith Milth (anti-drug NGO), Co-founder Gaash Lab (Mentoring)
Personal Interests	Badminton, swimming, blog (Journey with Bytes)

REFERENCES

[Prof. Chetan Arora](#) (IIT Delhi) [Dr. Mohit Jain](#) (Microsoft) [Prof. Raghu Krishnapuram](#) (IISc)